Tangential integration of the normal block

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Abstract

First geometric bridge after the normal block (Astra #17). For a continuous amplitude $\eta(v,u)$ on tangential×normal variables, a compact tangential set K and an integrable density q on K, the tangential average $\bar\eta(u) = \int_K q(v)\eta(v,u)\,dv$ is continuous (`continuous_tangentialAvg`), Fubini identifies the tangentially integrated dressed normal moment with the dressed moment of $\bar\eta$ (`integral_tangential_swap`), and hence

$$\frac{\int_K q(v)\int_{(0,1]^d}\eta(v,u)\,u^h e^{-\beta N u^{2k}}\,du\,dv}{N^{-\lambda}(\log N)^{|J|-1}} \longrightarrow \int_K q(v)\ \mathrm{amplitudeCoeff}(\eta(v,\cdot))\,dv$$

(`tangential_amplitude_tendsto`). Signed densities and amplitudes are allowed. Zero `sorry`/`axiom`.

1 The things formalised

```
def tangentialAvg (K) (q) (  : (Fin t →  ) × (Fin r →  ) →  ) (u) :   :=
   v in K, q v *  (v, u)
theorem continuous_tangentialAvg (K) (hKc : IsCompact K) (hK) (q) (hq : IntegrableOn q K)
    ( ) (h  : Continuous  ) : Continuous (tangentialAvg K q  )
theorem integrable_tangential_prod ... :
    Integrable (fun z => (q z.1 * w z.2) *   (z.1,   z.2))
      ((volume.restrict K).prod (volume.restrict S))
theorem integral_tangential_swap ... :
      v in K, q v *   u in S,   (v,   u) * w u
       =   u in S, (  v in K, q v *   (v,   u)) * w u
theorem tangential_amplitude_tendsto (d) (h k) (hk) (l  ) (hl) (h ) (hmin) (hatt)
    (K) (hKc) (hK) (q) (hq) (  : (Fin t →  ) × (Fin (d+1) →  ) →  ) (h  : Continuous  ) :
    Tendsto (fun N => (  v in K, q v *   x in unitBox (d + 1),
          (v, x) * ((  i, x i ^ h i) * exp (-(  * N *   i, x i ^ (2 * k i)))))
      / (N ^ (-l) * log N ^ (multCount (ratioExp h k) l - 1))) atTop
      (  (  v in K, q v * amplitudeCoeff h k l   (fun u =>   (v, u))))
```

2 Mathematics

Astra #17's "average first" route: the weight $u^h e^{-\beta N u^{2k}}$ does not depend on v, so Fubini moves the tangential integral inside, the amplitude theorem of unit 186 applies once to $\bar\eta$, and a second Fubini (with the face projection as the cube-valued map) turns $\mathrm{amplitudeCoeff}(\bar\eta)$ into $\int_K q\ \mathrm{amplitudeCoeff}(\eta(v,\cdot))$. Continuity of $\bar\eta$ is dominated continuity: near u_0 the amplitude is bounded on $K \times \overline{B}(u_0,1)$, so $C|q|$ dominates. Joint integrability on $K \times S$ uses the product integrability of $q(v)w(u)$ and the same compactness bound.

3 Paper-facing meaning

This is the deterministic chart-level precursor of the paper's tangential integration of the normal block against the density c_0 and the observable's normal data: for equal ratios the limit reduces to $\Gamma(\lambda)\beta^{-\lambda}/((d-1)!\prod 2k_i)\int_K q(v)\eta(v,0)\,dv$ (next unit), and for mixed ratios it retains the face integral in the nonminimal normal directions. The identification of $\int_K q(v)\ \cdot\ dv$ with $\int_{S_I} c_0|dv|$ requires the chart/partition-of-unity bridge, which remains outside the formalised boundary.

4 Lean notes

- `continuousAt_of_dominated` wants `x in x` measurability and bound; `Metric.closedBall_mem_nhds` supplies the neighbourhood on which the compact bound holds.
- For the product integrability, put the integrable factor first: `(hq.mul_prod hw).aestronglyMeasurable.mul (continuous measurable)` avoids any `AEStronglyMeasurable.fst` lemma.
- `Measure.prod_restrict` converts the product of restricted measures into the restriction to `K × S` so that `ae_restrict_iff'` applies.
- Definitions applied to a beta-redex (`amplitudeCoeff ... (fun u => (v, u))`) unfold to `(fun u => ...) (faceProj u)`: `unfold` then `beta_reduce` before `rw`.